

WAYNE COUNTY AP, OH, USA

WMO: 725216

Lat: 40.873N

Lon: 81.887W

Elev: 347

StdP: 97.23

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.6	-21.3	0.6	-17.2	-18.3	0.8	-13.9	11.9	-1.7	11.0	-1.2	2.5	280	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.2	22.7	29.8	22.2	28.6	21.5	24.3	28.8	23.5	27.8	22.7	26.7	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.3	26.9	22.2	17.6	26.0	21.3	16.7	25.0	74.9	29.1	71.6	27.7	68.3	26.6	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	8.9	8.0	-21.9	32.7	3.8	1.1	-24.6	33.6	-26.8	34.2	-29.0	34.8	-31.8	35.7
			-22.2	25.7	3.7	0.9	-24.8	26.4	-27.0	26.9	-29.1	27.4	-31.7	28.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-3.5	-2.1	3.0	9.8	15.7	20.3	22.0	20.7	17.3	11.2	5.1	-0.3	(6)
(7)		DBStd	10.22	6.81	6.40	6.19	5.12	4.62	3.51	2.80	2.73	3.85	4.86	5.10	5.70	(7)
(8)		HDD10.0	1591	418	342	231	65	7	0	0	0	1	43	161	323	(8)
(9)		HDD18.3	3389	676	573	474	258	109	21	4	9	63	226	397	579	(9)
(10)		CDD10.0	1593	1	3	15	60	183	309	371	333	221	81	15	2	(10)
(11)		CDD18.3	348	0	0	0	3	27	80	116	84	33	6	0	0	(11)
(12)		CDH23.3	2997	0	0	4	42	266	656	974	676	324	55	1	0	(12)
(13)		CDH26.7	789	0	0	0	4	57	187	287	158	86	9	0	0	(13)
(14)	Wind	WSAvg	3.4	4.5	4.3	4.1	4.2	3.4	3.1	2.5	2.1	2.3	3.0	3.7	4.1	(14)
(15)	Precipitation	PrecAvg	941	57	47	68	85	102	95	100	99	84	63	75	65	(15)
(16)		PrecMax	1354	166	118	164	160	211	221	325	240	217	149	239	176	(16)
(17)		PrecMin	572	14	5	20	20	27	17	18	12	14	2	13	16	(17)
(18)		PrecStd	168	32	22	28	36	43	48	64	45	47	34	39	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	18.1	22.8	27.2	30.2	32.3	32.7	31.8	31.4	27.9	21.1	15.9	(19)
(20)		0.4%	MCWB	12.9	13.3	15.6	17.7	20.8	22.1	23.8	23.3	22.3	20.6	14.6	12.9	(20)
(21)		2%	DB	12.4	13.4	18.8	24.2	28.2	30.6	31.1	30.0	29.1	24.6	17.9	12.9	(21)
(22)		2%	MCWB	10.6	10.7	13.5	15.7	19.6	21.8	23.4	22.8	21.5	17.7	13.5	10.5	(22)
(23)		5%	DB	9.0	10.2	16.0	22.0	26.6	28.8	29.6	28.4	27.2	22.2	16.0	10.6	(23)
(24)		5%	MCWB	7.2	7.8	11.6	14.6	19.0	21.1	22.6	22.0	20.6	16.5	12.2	8.6	(24)
(25)		10%	DB	6.0	6.8	13.0	19.0	24.2	27.3	28.1	27.1	24.9	19.6	13.4	7.8	(25)
(26)		10%	MCWB	4.2	4.4	9.8	13.1	17.9	20.5	21.7	21.2	19.1	15.7	10.3	6.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	14.2	16.5	19.2	22.7	24.1	25.7	25.1	24.2	21.8	16.7	13.5	(27)
(28)		0.4%	MADB	15.2	17.6	22.0	24.7	28.2	29.4	30.7	29.6	29.1	26.7	19.8	15.1	(28)
(29)		2%	WB	10.9	11.1	14.3	17.2	21.2	23.1	24.6	24.0	22.8	19.7	14.4	11.2	(29)
(30)		2%	MADB	12.4	12.8	18.0	22.4	26.1	28.0	29.2	28.1	27.0	22.6	16.9	12.6	(30)
(31)		5%	WB	7.7	7.8	12.4	15.7	20.0	22.3	23.7	23.0	21.6	17.7	12.6	8.8	(31)
(32)		5%	MADB	8.9	10.0	15.4	20.4	24.7	27.1	27.9	26.6	25.4	20.6	15.2	10.3	(32)
(33)		10%	WB	4.2	4.4	10.0	14.0	18.9	21.4	22.8	22.2	20.4	15.8	10.6	6.2	(33)
(34)		10%	MADB	5.8	6.6	13.0	18.1	23.0	25.7	26.7	25.5	23.7	19.1	13.3	8.0	(34)
(35)	Mean Daily Temperature Range	MDBR	8.2	8.7	9.7	11.8	11.8	10.9	11.0	11.4	12.4	11.3	9.3	7.2	(35)	
(36)		5% DB	MADB	10.4	12.2	14.0	15.3	14.1	12.9	12.3	12.8	14.5	14.7	13.1	10.2	(36)
(37)		5% DB	MCWBR	9.0	9.7	9.2	8.9	7.1	6.0	5.7	6.6	7.6	8.4	9.1	8.7	(37)
(38)		5% WB	MADB	9.6	11.7	12.8	13.1	11.9	11.0	10.3	10.3	11.9	11.9	11.3	9.9	(38)
(39)		5% WB	MCWBR	9.1	10.0	9.0	8.7	6.7	5.7	5.2	5.9	6.7	7.7	8.8	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.328	0.351	0.386	0.422	0.444	0.444	0.431	0.379	0.360	0.320	0.303	(40)	
(41)		taud	2.424	2.329	2.351	2.280	2.209	2.186	2.205	2.227	2.350	2.396	2.475	2.451	(41)	
(42)		Ebn at Noon	858	879	898	885	857	836	832	832	856	833	828	823	(42)	
(43)		Edh at Noon	80	102	112	129	141	145	141	134	111	95	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.46	2.32	3.33	4.54	5.40	5.91	5.90	5.33	4.29	2.83	1.84	1.23	(44)	
(45)		RadStd	0.13	0.30	0.33	0.42	0.45	0.44	0.38	0.26	0.39	0.25	0.27	0.14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	(46)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	N/A	(47)

Nomenclature: See separate page